

Bence Jordan Danko

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EDUCATION

San Jose State University
M.Sc. in Applied Data Intelligence
San Jose State University
B.Sc. in Computer Science

2024 – Present
San Jose, USA
2020 – 2024
San Jose, USA

PROJECTS

Edge Vision Smart Elevators: Privacy-Preserving Headcount Analysis

<https://github.com/bencejdanko/Edge-AI-People-Counting-NuMaker-X-M55M1D>

- Conducted applied research for Nuvoton's NuMicro M55M1 microcontroller, architecting a confined person-recognition pipeline achieving 15 FPS within strict 1.5MB SRAM constraints.
- Designed and deployed an ultra-efficient FOMO (Faster Objects, More Objects) model for microcontrollers, reducing the memory footprint to less than 45KB, a 98% reduction compared a 2.7MB YOLOv8n approach.
- Engineered an INT8-quantized object detection pipeline optimized via the Vera compiler to run directly on the hardware's Arm Ethos-U55 NPU.
- Developed custom C++ firmware for Arm Cortex-M55 architecture, designing on-chip UDP server and FOMO inference handlers for real-time microcontroller communication and data transfer for realtime AI inference.
- Utilized Keil IDE, hardware debuggers, and stack trace analysis to isolate complex runtime bugs, optimize memory allocation, and resolve critical stack/heap behavior issues.
- Assembled a proprietary data collection setup using a Raspberry Pi Zero W and a 160-degree FOV camera to capture real-world data in confined elevator environments.
- Curated a specialized dataset of 13,000+ images from Roboflow, subsequently formatting and distributing it for public use on Hugging Face.
- Collaborated directly with Nuvoton hardware engineers to validate end-to-end system performance and prepare the project for a live demonstration at the 2026 Sensors Converge conference.

Segmentation/OCR Multimodal API Agent for Grazen AI

<https://warehouse.openwear.ai>

- Collaborated with startup researchers to develop a multimodal API agent for warehouse logistics to automate inventory audits.
- Utilizes SAM 3 for object localization and GLM-OCR for text extraction to reduce manual labor and inventory losses.
- Implemented an orchestration pipeline that integrates SAM 3 segmentation with LLM-based filtering, allowing for dynamic re-prompting and precise localization of front-row product stacks.
- Built a comprehensive data extraction service that combines bounding box analysis and OCR processing to synthesize structured inventory reports, including product names, instance counts, and quantity validation.
- Engineered a dashboard for displaying real-time segmented crops, detected product counts, and extracted textual labels for warehouse management validation.

Mobilint MLA100 NPU Computer Vision Applications

<https://github.com/bencejdanko/image-restoration-convolution-mla100-optq-qat>

- Worked with Mobilint hardware and compiler engineers to evaluate computer vision applications for the MLA100 NPU.
- Developed and optimized a real-time image restoration pipeline for the NPU, achieving <6ms latencies for 256x256 images.
- Replaced a high-latency ResNet baseline (400ms+) with a custom 0.2M parameter Convolutional Network by optimizing kernel launches and restricting feature maps to 64 to minimize hardware-induced latency.
- Orchestrated large-scale training and data processing jobs using modal, scaling cloud GPU resources to conduct progressive training cycles with Cosine Annealing and mixed precision.
- Evaluated and compiled ImageNet classification models for the MLA100, comparing inference throughput against Intel AI Boost and achieving 2.7x performance gains.